

Packet Tracer – Spanning Tree Protocol (STP) Loop Prevention and Redundancy

Objectives

Part 1: Build a Redundant Switched Topology

Part 2: Configure Basic IP Addressing

Part 3: Verify Normal Connectivity

Part 4: Observe STP Operation

Part 5: Configure the Root Bridge

Part 6: Test Loop Prevention

Part 7: Verify Redundancy and Failover

Background / Scenario

You are the network administrator of a company that requires high availability between switches.

To provide redundancy, multiple links have been installed between switches. However, redundant links can create Layer 2 switching loops, causing:

- Broadcast storms
- MAC address table instability
- Duplicate frame forwarding

To prevent these problems, switches use **Spanning Tree Protocol (STP)**.

In this lab, you will:

- Build a redundant switching topology
- Observe STP elect a Root Bridge
- Identify blocked ports
- Test network redundancy
- Simulate link failures
- Verify automatic failover

This lab demonstrates how STP prevents loops while maintaining backup paths for network resiliency.

Instructions

Part 1: Build the Topology

Step 1: Add Devices

Add:

- 3 Switches (2960)
- 2 PCs

Rename:

- Switch0 → SW1
 - Switch1 → SW2
 - Switch2 → SW3
 - PC0 → PC-A
 - PC1 → PC-B
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Step 2: Connect the Devices

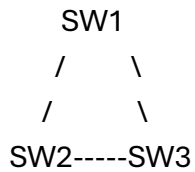
Use Copper Straight-Through cables.

Device Interface Device Interface

PC-A	Fa0	SW1	Fa0/1
PC-B	Fa0	SW3	Fa0/1
SW1	Fa0/23	SW2	Fa0/23
SW2	Fa0/24	SW3	Fa0/24
SW1	Fa0/22	SW3	Fa0/22

The switches should form a triangle.

Example:



This topology contains a Layer 2 loop.

Without STP, the network would become unstable.

Part 2: Configure Basic IP Addressing

Since all devices belong to the same VLAN, use one subnet.

Step 1: Configure PC-A

Desktop → IP Configuration

Enter:

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: Leave Blank

Step 2: Configure PC-B

Desktop → IP Configuration

Enter:

IP Address: 192.168.1.20

Subnet Mask: 255.255.255.0

Default Gateway: Leave Blank

Part 3: Verify Connectivity

From PC-A:

ping 192.168.1.20

Expected:

Reply from 192.168.1.20

4 successful replies.

From PC-B:

ping 192.168.1.10

Expected:

Reply from 192.168.1.10

At this stage communication should work normally.

Part 4: Observe STP Operation

Because the topology contains a loop, STP automatically activates.

Step 1: Check STP Status

On SW1:

show spanning-tree

Repeat on:

SW2

SW3

Expected:

One switch becomes the Root Bridge.

Example:

This bridge is the root

or

Root ID

information displayed.

Step 2: Identify Blocked Ports

On one switch, a port should be in:

BLK

or

Blocking

state.

Expected:

One redundant link is blocked

This prevents switching loops.

Part 5: Configure a Root Bridge

By default, the Root Bridge is selected using the lowest Bridge ID.

You will manually force SW1 to become the Root Bridge.

Step 1: Configure SW1

enable

configure terminal

spanning-tree vlan 1 root primary

end

Step 2: Verify Root Bridge

On SW1:

show spanning-tree

Expected:

This bridge is the root

On SW2 and SW3:

show spanning-tree

Expected:

Root ID Priority

showing SW1 as Root.

Part 6: Test Loop Prevention

STP should keep one redundant path blocked.

Step 1: Verify Port Roles

On SW2 or SW3:

show spanning-tree

Expected port roles:

Root Port

Designated Port

Alternate Port

One port should remain blocked.

Step 2: Confirm Network Connectivity

From PC-A:

ping 192.168.1.20

Expected:

Reply received

Even though one path is blocked, connectivity remains available.

Part 7: Test Redundancy and Failover

Now simulate a network failure.

Step 1: Disconnect Active Link

Remove the cable between:

SW1 Fa0/23

and

SW2 Fa0/23

or shut it down:

```
interface fa0/23
```

```
shutdown
```

Step 2: Wait for STP Recalculation

Wait approximately:

30–50 seconds

for classic STP convergence.

Step 3: Verify Connectivity Again

From PC-A:

```
ping 192.168.1.20
```

Expected:

Reply received

Connectivity should return through the previously blocked backup path.

Step 4: Verify New STP Topology

On SW2:

show spanning-tree

Expected:

Previously blocked port is now forwarding

STP has activated the backup path.

Verification Commands

Verify Root Bridge

show spanning-tree

Verify Interface Status

show interfaces status

Verify MAC Address Learning

show mac address-table

Verify Port Roles

show spanning-tree vlan 1

Expected roles:

Root Port

Designated Port

Alternate Port

Troubleshooting Tips

If connectivity fails:

✓ Verify all switch links are connected

✓ Ensure all ports are in VLAN 1

✓ Verify interfaces are not shutdown

show ip interface brief

✓ Verify STP is running

show spanning-tree

✓ Wait for STP convergence

✓ Check that at least one path remains active

✓ Verify PCs are in the same subnet

What You Learned

- Why Layer 2 loops are dangerous
 - How STP prevents broadcast storms
 - How Root Bridge election works
 - How STP selects Root, Designated, and Alternate ports
 - How blocked ports prevent switching loops
 - How redundant links provide fault tolerance
 - How STP automatically recovers after a link failure
 - How to verify STP operation using Cisco IOS commands
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Challenge Extensions (Optional)

Challenge 1: Make SW2 the Root Bridge

Configure SW2:

spanning-tree vlan 1 root primary

Verify that Root Bridge election changes.

Challenge 2: Configure Secondary Root

On SW3:

spanning-tree vlan 1 root secondary

Verify the backup root role.

Challenge 3: Enable PortFast

On PC-facing ports:

interface fa0/1

spanning-tree portfast

Observe faster PC connectivity.

Challenge 4: Observe STP in Simulation Mode

Switch Packet Tracer to:

Simulation Mode

Generate:

- ARP traffic
- Ping traffic
- Broadcast traffic

Observe STP BPDUs exchanged between switches.

Challenge 5: Compare STP and RSTP

Change spanning-tree mode:

spanning-tree mode rapid-pvst

Disconnect a link and compare failover time with traditional STP.